Lecture plan · for the lecturer

Notional machines and tracing

Programming · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Define notional machines.
- Recognise misconceptions.
- Design diagnostic questions.

Activity: Write diagnostic questions

1. Warm-up (5 min): Show the three-line trace and ask everyone to vote before discussing.
2. Teach (10 min): Present five misconceptions from Sorva's catalogue with examples.
3. Write (25 min): Pairs write one tracing question whose distractors each match a misconception.
4. Swap (15 min): Pairs swap questions and check which model each wrong answer reveals.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which misconception surprised you?
- What makes a distractor diagnostic?
- Where will you draw memory?

Exit ticket

- Write one distractor and the misconception it detects.

Try it in your next lecture

Start your next lecture with three tracing questions whose distractors each match a known misconception; students vote on their phones, then redraw memory for the most-missed question.

Seminar sheet · for students

Notional machines and tracing

Programming · University · Lecture 3

Name _____ Class _____ Date _____

1 **A student answers 7 7. Which misconception does that reveal?**

◯ Assignment creates a lasting link ◯ Print shows the last line only ◯ Variables can't change

2 **Trace the swap.**

x = 1, y = 2, then t = x, x = y, y = t. Now x is _____ and y is _____ .

Word bank: 2 · 0 · 3 · 1

3 **Which are documented novice misconceptions? Choose all that apply.**

☐ Assignment links variables permanently ☐ A variable can hold several values at once

☐ Each call has its own local variables ☐ The computer infers what the programmer meant

4 **A student who can write correct code on a quiz necessarily has a viable model of assignment.**

◯ True ◯ False

5 **Match each wrong answer to the misconception it reveals.**

b = 5; a = b; b = 7 → a is 7	• • • •	The computer knows intent
x = 3; x = 4 → x is 3 and 4	• • • •	Definition is execution
A defined function runs without being called	• • • •	A variable holds history
Misspelt name still works	• • • •	Assignment as a link

✎ **Which misconception do you suspect is common in your course but have never measured?**